MLDM Assignment 9

Detailed results for Jandson Ribeiro

Course	**Machine Learning and Data Mining WS 2022/23 (copy)**
Test	**MLDM Assignment 9**

This are your test results

Knowledge Questions

SVM

Status	Seen but not answered	
Your score	0 / 10	0%

Response

For the statements about Support Vector Machine given below, select if each statement is right / wrong.

Note :

Correct answer : +2 points

Incorrect answer : -1 point

Unanswered	Right	Wrong	
⊙	○	○	A sensible hyperplane seperating two classes , should leave minimum margin on either sides to reduce error when presented with unknown data.
⊙	○	○	While using polar coordinates to separate non-linearly separable classes, the distance from a reference point and angle of each point with the other point is used instead of the actual coordinates of the points in the given dimension.
⊙	○	○	The value of Slack variable is greater than 0 when the sample falls on the wrong side.
⊙	○	○	In a SVM, a 2 dimensional plane becomes the hyperplane that classifies the datapoints when the number of input features is 2.
⊙	○	○	If we retrain SVM only with the support vectors, we would obtain the same hyperplane.

Feed forward neural network

Status	Not started	
Your score	0 / 10	0%

Response

For the statements about neural networks given below, select if each statement is right / wrong.

Note :

Correct answer : +2 points

Incorrect answer : -1 point

Unanswered	Right	Wrong	
⊙	○	○	While adjusting synaptic weights in a neural network using Backpropagation, the convergence speed depends on the value of the bias.
⊙	○	○	The training of neural networks is very sensitive to initial weights.
⊙	○	○	While using Backpropagation, in order to minimise the cost function, the derivatives of this cost function w.r.t all controlling variables have to be computed.
⊙	○	○	In a 2 layer Perceptron, the first phase is to transform the non linearly separable problem into a linearly separable problem.
⊙	○	○	In Machine learning in general and in Neural networks in particular, maximising the log-likelihood is different than minimizing the cost function.

Activation function

Status	Not started		
Your score	0 / 6.5		0%

Response

The activation function introduces non linearity in the model.

a) What is the necessity of the non-linearity in Multi-Layer perceptron? How does it affect the Hidden Layer?

b) Why do we need hidden layers in Multi-Layer perception?

0 word

Learning

Status	Not started		
Your score	0 / 6		0%

Response

Explain in your own words:

- How backpropagation works.
- The role of Gradient descent.
- The difference between Gradient descent and stochastic gradient descent.

0 word

Practice

Forward pass

Status	Not started		
Your score	0 / 15		0%

Response

Given below is a simple 2-level neural network.

Simple neural network

This network consists of 2 input neurons, 2 hidden neurons (h1, h2) and 1 output neuron. The hidden neurons and output neurons include a sigmoid activation function. Input , target Output and bias are constant.

Calculate the feed forward pass by processing the inputs through the neural network with given weights. Use the following loss function

$E_{\text{total}} = \sum(\text{target} - \text{output})^2$

Total net input for h1 :

Output of h1 :

Input of h2 :

Final Output received :

Total Error in the output neuron :

Backward pass

Status	Not started	
Your score	0 / 20	0%

Response

On the neural network from the previous task, apply Gradient descent and find how w_4 affects the total error. The neural netwrok image is given below again for reference.

In order to identify how much change in w_4 affects the total error, find the following values.

net_{o1} is the input of output neuron: $g(h_{-1})$
out_{o1} is the output of the output neuron: $f(g(h_{-1}))$.

Give your answer with 3 decimal values.

$\frac{\varphi E_{\text{total}}}{\varphi \text{out}_{o1}}$ =

$\frac{\varphi \text{out}_{O1}}{\varphi \text{net}_{o1}}$ =

$\frac{\varphi \text{net}_{o1}}{\varphi W_4}$ =

$\frac{\varphi E_{\text{total}}}{\varphi W_4}$ =

Update value of w_4 once with a learning rate of $\eta = 0.5$

Updated w_4 =

Map higher dimension

Status	Not started	
Your score	0 / 7.5	0%

Response

Transform the following data points into a higher dimensional feature space to find a hyperplane separating the 2 classes clearly using the kernel trick.

Use the given Kernel for mapping.

Map into higher Dimension

Give, the new datapoints after mapping into the higher dimension.

(0,6) --> x : y : z :

(1,5) --> x : y : z :

(3,3) --> x : y : z :

(4,2) --> x : y : z :

(6,0) --> x : y : z :

Note:

If there are decimal values, give decimal values with 2 places

Each correct answer gets 0.5 point

Programming

Feed forward algorithm

Status	Not started		
Your score	0 / 25		0%

Response

Write python functions to perform the following tasks in a neural network .

1. Feed forward Algorithm.

   def feed_forward():

2. Sigmoid Activation Function.

   def sig_activation():

3. Output Function that calls Functions 1 and 2 .

   def call_output():

Paste your code in the answer box below

Note :

In each of the above functions accept necessary parameters. Use the given Function names while writing the functions.

Do not import any advanced packages for this task. Use only primary packages.

0 word